EE3032 Introduction to VLSI Design—Homework 1

1. **MOS Theory** (4 points)
   Fig. 1 shows two CMOS inverters. Assume that the body effect can be neglected.

   (a) Fig. 1(a) shows an inverter with four transistors. Identify the transistor(s) which cannot operate in the saturation region. Explain your answer. (2 points)

   (b) Assume that the transistor size of M1∼M4 is (W/L). Determine the transistor size of N1 and N2 shown in Fig. 1(b) such that the two inverters have the same characteristics. Explain your answer. (2 points)

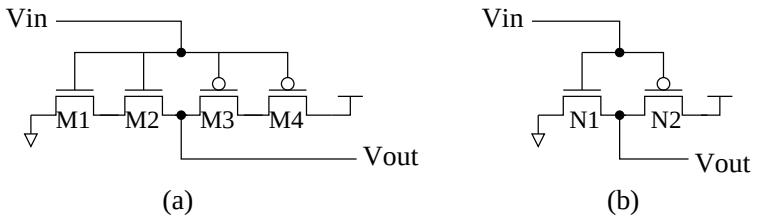

Figure 1: CMOS inverters

2. **Multi-V_t Design** (4 points)
   Fig. 2(a) shows a simple combinational logic. Assume that the delay of each logic gate is 1 unit delay if all the transistors used to realize the logic gates are low V_t transistors. If we want to reduce the subthreshold conduction power of the circuit using multiple-V_t process (i.e., you have high-V_t and low-V_t transistors) without sacrificing the performance of the circuit, please draw the transistor-level schematic of the combinational circuit with multiple-V_t transistors using the symbols shown in Fig. 2(b).

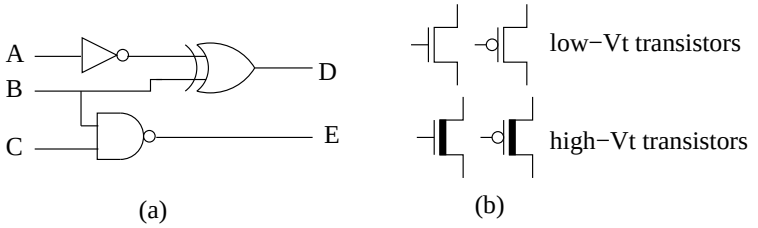

Figure 2: (a) A simple CMOS logic (b) Symbols of transistors

3. **Multiplexer-Based Design** (3 points)
   According to *Shannon's expansion theorem*, a Boolean function with n variables can be expanded on any one of variable x_i. That is, $f(x_1, x_2, \ldots, x_i, \ldots, x_n) = x_i \cdot f(x_1, x_2, \ldots, 1, \ldots, x_n) + \bar{x}_i \cdot f(x_1, x_2, \ldots, 0, \ldots, x_n)$. This implies that we can realize a Boolean function using multiplexers. Fig. 3 shows an example of a two-input AND gate realized by a 2-to-1 multiplexer. Use 2-to-1 multiplexers to implement $f(x_1, x_2, x_3) = \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3$.

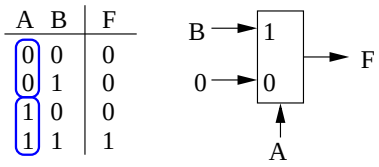

Figure 3: Using multiplexer to realize a two-input AND gate

4. **Static CMOS Logic** (4 points)
   Realize each of the following Boolean functions using static CMOS logic: (a) $f(w,x,y,z) = w \cdot (x+y) + y \cdot z$ (b) $f(x,y,z) = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}$. Suppose that only true literals are available. You should show the transistor diagram.